

An evolutionary game-theoretic approach to the strategies of community members under Joint Forest Management regime

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Abstract

Joint Forest Management (JFM) has been analyzed using an evolutionary-game-theoretic approach. The interactions between the different groups of a community, for forest use under state regime and JFM regime, are modeled as n -person asymmetric games, and the concepts of evolutionary stable strategies (ESS) and asymptotically stable states (ASS) are used to understand the variations in the outcomes of JFM program. The n -person game of forest use under the state regime has a unique Nash equilibrium in which the defectors or lawbreakers will continue to harvest forest resources illegally until the net returns from harvests become negative. The n -person forest resource use game under JFM regime has many Nash equilibriums, but has only one sub-game perfect defection equilibrium. However, the n -person game for JFM regime has four evolutionary strategy equilibriums: cooperators (C) equilibrium, defectors (D) equilibrium, defectors–enforcers (D–E) equilibrium and cooperators–enforcers (C–E) equilibrium, but has only two asymptotically stable (C–E and D–E) equilibriums. Implications of these results are discussed, and a need to enhance evolutionary game-theoretic formulation of JFM is highlighted.

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1. Introduction

In recent years, many economists, including Arthur (1994), Bowles (2004) Colander (2000), Friedman (2004), Kahneman and Tversky (2000), Kant (2005), McFadden (1999) and Smith (2000), have challenged

the foundations of the rational economic agent which is commonly known as *Chicago man*. In the words of Sen (1977), the rational economic agent is close to being a “social moron” or a “rational fool” while according to Hegel (1967) the agent is a “mindless individual.” Recently, the *Chicago man* has become an endangered species, his maximum range has been severely restricted, and he is not safe even in markets for concrete goods, which was his prime habitat (McFadden, 1999). Thaler (2000) predicted that *Homo economicus* will evolve into *Homo sapiens* who will have characteristics of lower IQ,

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slow learning, human cognition and more emotions. As a result of these challenges by various scholars, economic decisions are now routinely analyzed by assuming that human beings follow some process of selection, based on procedural rationality—perhaps the tendency of people to learn from their past actions and to imitate their more successful counter parts. The procedural rationality makes maximization to be a tiny subset of the vast repertoire of possible human behaviors (Samuelson, 2002).

Non-cooperative game theory is one of the standard tools of economic analysis, and the initial focus of the game-theoretic models was also on the *Chicago man*. However, the Chicago-man-based game-theoretic models have suffered from various limitations (Samuelson, 2002). First, an assumption of perfect rationality in the game-theoretic models, representing an approximation of an actual interaction, is not realistic. Second, rationality-based criteria for choosing among multiple Nash equilibrium produced alternative “equilibrium refinements” to prompt despair at the thought of ever choosing one as the “right” concept. Finally, these models readily motivated one of the requirements of Nash equilibrium—players choose best responses to their beliefs about others’ responses—but provided no evidence that these beliefs are correct. As a result, in the recent years, emphasis of game-theoretic models has shifted from the *Chicago man* to evolutionary models which have brought the learning and imitation portion of human behavior to the forefront of economic analysis.

The common theme in “evolutionary games” is a dynamic process, which describes how players adapt their behavior over the course of repeated plays of a game. The dynamic process provides the coordination device that brings beliefs into line with behavior through the individual learning process; it provides a context for play that may be useful in assessing multiple equilibriums; and views equilibrium as the outcome of an adjustment process, a realistic version of human interactions (Fudenberg and Levine, 1997; Samuelson, 1997; Fudenberg and Maskin, 1990). The main assumption of evolutionary game theory is that large populations of “procedurally rational” players learn and imitate, copy successful strategies of others and gradually discard unsuccessful strategies (Cressman et al., 1998).

The evolutionary game theory, due to its features of learning, imitation and selection processes followed by

economic agents, has become a valuable tool for economic analyses. In a short period of a decade or so, it has been applied to a wide range of areas including evolution of social norms (Axelrod, 1984, 1986; Hofbauer and Sigmund, 1988; Binmore and Samuelson, 1994), evolution of community norms in the case of common pool resource use (Sethi and Somanathan, 1996), dynamics of technology policy (Metcalf, 1994), dynamics of crime (Cressman et al., 1998) and dynamics of financial innovations (Bettzuge and Hens, 2001). In this paper, we apply evolutionary game theory to analyze the process of co-management or joint management of forest resources by local communities and the state.

In developing countries, communities living in and around forests are highly dependent on these forests for their day-to-day needs of fuel wood, small timber, fodder and other non-timber forest products. Hence, the exclusion of local communities from forest use/management is almost impossible, and exclusionary policies have been the main reason of deforestation and forest degradation in these areas (Pacheco, 2004). In the last two decades, many developing countries, including Cambodia (Marschke and Nong, 2003), China (Xu et al., 2004), Ethiopia (Gebremedhin et al., 2003), India (Kant, in press, 2000, Kant and Berry, 2001), Mexico (Klooster, 2000; Munoz-Pina et al., 2002) and Nepal (Mathema, 2004) have tried to resolve this problem by involving local communities in forest management, and these programs are known as co-management, community-based forest management, or joint forest management (JFM)¹. The two main features of these programs are (i) forest managers, normally state agencies, seek the cooperation of local communities in forest protection and forest management; and (ii) local communities are ensured, by forest managers, of a share in the final harvest of timber in addition to the annual harvest of non-timber forest products and wages for their forest protection and management work (Shahi and Kant, 2005). Researchers and forest managers have reported wide spatial and temporal variations in the outcomes of these programs across different communities and dif-

¹ Co-management programs of natural resources are not limited to developing countries. Some co-management programs are also reported from Australia (Ananda and Herath, 2003), Canada (Innes, 2003; Sheppard and Meitner, 2005) and Finland (Leskinen, 2004).

ferent types of forests (Baland and Platteau, 2003), but the economic rational behind these variations of outcomes has not been very well established.

Shahi and Kant (2005) used an evolutionary game-theoretic approach to model the interactions for JFM, between local communities and state agencies (forest managers). They formulated an asymmetric bi-matrix game to characterize these interactions, and used the concept of evolutionary stable strategies (ESS) to understand the spatial and temporal variations in the outcomes of JFM programs. However, the results of their paper are limited to the communities where all members of the community follow the same strategy or, from the economic strategies perspective, the communities are perfectly homogenous which may not be a realistic situation in many communities. Shahi and Kant (2005) recognized this limitation and emphasized the need to extend their work to heterogeneous communities, and in this paper, we do that.

We first formulate an n -person game for a community, comprising of law abiders (or cooperators) and law breakers (or defectors), forest use under state regime, and demonstrate that this game has a unique Nash equilibrium in which defectors earn a higher payoff and as such they exploit the forest resource, leading to its degradation and even complete extinction. Next, we formulate an n -person game for a community, comprising of cooperators (C), defectors (D) and enforcers (E), forest use under JFM regime, and demonstrate that this game has many Nash equilibriums, but has a unique sub-game perfect defection equilibrium. Finally, we examine the evolutionary dynamics of the game for JFM regime and demonstrate that this game has four evolutionary strategy equilibriums: cooperators (C) equilibrium, defectors (D) equilibrium, defectors–enforcers (D–E) equilibrium and cooperators–enforcers (C–E) equilibrium; but it has only two asymptotically stable (C–E and D–E) equilibriums.

In the next section, we provide an overview of the literature related to joint forest management and game theory. In Section 3, we discuss an n -person community forest use game under state regime while an n -person community forest use game for JFM regime is discussed in Section 4. In Section 5, we discuss the replicator dynamics of the n -person game for JFM regime. Finally, we conclude with some implications of our results and need to enhance evolutionary game-

theoretic formulation of JFM. In this paper, the JFM model has been developed with particular reference to the state of JFM in India. Nevertheless, the broad outcomes should be applicable to most co-management situations around the world.

2. Joint forest management, economics and game-theoretic models

The economics of forest resource management is still controlled by the paradigm of the *Chicago man*, and as a result procedural rationality aspects of human behavior are almost totally missing from forest economics literature except for a few examples such as Kant (2003) and Kant and Berry (2005). However, joint forest management or co-management has been accepted as one of the possible economically efficient regimes, and many economists, such as Hill and Shields (1998), Ligon and Narain (1999), Kumar (2002), Kant et al. (1996), Richards et al. (2003) and Ananda and Herath (2003) have conducted economic analysis of JFM using the concepts of perfect rationality. Kant (in press, 2000) and Kant and Berry (2001) extended the economic analysis of JFM to the new institutional analysis framework by incorporating a transaction function specified in terms of two socio-economic factors—dependence and heterogeneity—of local community.

Use of game-theoretic models in forest management is really scarce, and there are only few papers on the subject. Kant and Nautiyal (1994) used cooperative game theory to suggest a mechanism, based on community's and government's fixed threat points, to fix the community share in the final timber harvest under the JFM program in India. Angelsen (2001) explored possible strategic interactions between the state and local community in games of tropical forestland appropriation, and concluded that the game structure (Cournot or Stackelberg) is important for total deforestation. Lise (2001) proposed a stepwise procedure to derive the payoffs and actions in a non-cooperative game, and applied it to a village under JFM in India.

Game-theoretic models, however, have been used extensively to analyze common pool resource problems, which include specific cases of forest management. Ostrom et al. (1994) used both experimental and field data to test models, based on the theory of n -person finitely repeated games of human behavior in

common pool resource situations. Apesteguia (2001) analyzed the effect of availability of information about payoff structure on the behavior of players in a common pool resource game. Rotillon and Tazdait (2003) studied the question of cooperation in the presence of global environmental commons and found that a possible solution for the tragedy of commons does not necessarily imply cooperative behavior of all the countries involved. Caputo and Lueck (2003) developed a model of a common property contract based on differential game theory to examine the incentives of individual users of a common resource. The concepts of evolutionary game theory² have even been used by Sethi and Somanathan (1996) to explain the evolution of social norms in the context of common pool resource use.

Generally, common pool resource use situations, at least the situations analyzed in game-theoretic literature, are different than the situations of JFM or co-management of forest resources. In the common pool resource situations, there is only one population of local community comprising of cooperators, defectors and in some cases enforcers, and the payoffs of these three categories of agents are basically determined within the community, while in JFM situations, the state agencies (forest managers) play a critical role in the payoffs of the three groups of agents. Hence, in this paper, we formulate and analyze a game situation, using the concepts of evolutionary game theory, which is quite different than that of the games of common pool resources.

3. *n*-person game of forest resource use under state regime

In developing countries, members of local communities living in surrounding areas of forest resources are dependent on these resources for their daily subsistence needs such as fuel wood and fodder; however, they do not have free access to these resources which are gen-

erally owned and managed by the state. Forest managers (state employees) are responsible for forest management, as well as enforcing property rights. However, forest managers cannot ensure complete exclusion of local communities from using the resource due to large physical boundaries of the resource, proximity of communities to the resource, and limited financial and human resources at the disposal of forest departments. Due to the non-availability of alternate resources for their livelihood, some of the members of the community use the government owned forests illegally for collection of fuel wood and fodder and for their cattle to graze. Some members of the community even resort to illegal removal of timber and non-timber forest products for their subsistence income while some members also resort to illegal harvesting for other needs. Hence, in every community living close to forests, two groups of people can be found—one law abiding and the other using the forests illegally.

Suppose R_i is the annual payoff (net of labor cost) per unit of effort from illegal removals of forest produce from forests, R_f is the annual fine (value of forest produce and punishment for theft) paid by an illegal harvester if he is caught by a forest manager, and b is the probability of being caught by a forest manager which is normally very small. The net annual payoff of a person, who removes illegal forest produce from the resource is $(eR_i - bR_f)$, where e is the effort exerted by the person for illegal removal of forest produce.

The annual payoff per unit effort (R_i) depends on the growing stock (or value) of timber and non-timber forest products, which is a function of the site index and the age of forest resource. Suppose R_{is} and R_{it} are the first order partial derivatives of R_i with respect to the site index and the age of the forest, respectively, then $R_{is} > 0$ and $R_{it} > 0$, i.e., the annual payoff from illegal removal per unit effort increases with the site quality and age of the forest resource. For the sake of simplicity, we assume that there are only two site qualities, a good and a poor site quality, and these site qualities support a high quality and a poor quality forest, respectively. In the case of a high quality forest, R_i is generally high for all age classes, while in the case of a poor quality forest R_i will be high only in the case of old and matured trees.

In either type of forests, of high or poor quality, illegal harvesters will continue to remove forest

² We do not intend to review the whole literature on evolutionary game theory. However, the interested readers may like to refer to some of the recent literature on the topic such as Taylor and Jonker (1978), Zeeman (1979), Samuelson and Zhang (1992), Ellison (1993), Kandori et al. (1993), Binmore and Samuelson (1994), Sethi (1996, 2000), Sethi and Somanathan (2001), Binmore and Samuelson (2001), Samuelson and Swinkels (2003), Blume (2003) and Schuster and Perelberg (2004).

produce until their net annual payoff from illegal harvest is zero. Since, in the case of high quality forests, R_i is very high, illegal harvesters will continue to harvest timber and other forest produce illegally until the forest is degraded to a stage where the net returns from these activities becomes zero. In the case of high quality forests, it may take a longer time due to the high growing stock and high quality of forest products, but the final outcome of this process, irrespective of time, is degraded and denuded forests. The same phenomenon will take place in poor quality forests, but they may be reduced to the degraded and denuded state in a much shorter time.

A relationship of a local community with the forest, which is under state regime, can be described by an n -person game where n is the number of people or a household (economic agents) in the community. All the economic agents belong either to the law abiding group or to the illegal harvesters group. The net annual payoff from forests to the law abiding agents is always zero while the net pay off from forests to the illegal harvester member is $(eR_i - bR_f)$. In this game, there is a unique Nash Equilibrium in which the lawbreakers would like to choose an effort level that maximizes their payoff from the resource. Even if the number of lawbreakers in the community is very small, such Nash equilibrium will continuously lead to the depletion of the resource and if the resource is already degraded, this equilibrium may possibly lead to its extinction over a period of time. Such an outcome can only be avoided either by increasing R_f or b . JFM is an effort in this direction.

4. n -person game of forest resource use under joint forest management regime

As stated in the Introduction, governments in many developing countries have realized these situations, and have started co-management or Joint Forest Management (JFM) with the involvement of local communities in the protection and management of forest resources. Under the JFM program, the government signs an agreement with a community, which outlines the rights and duties of the government and the community, agreed upon by each side. As per these agreements of JFM in India, the community has to protect the forest until their maturity and confine grazing to certain pre-selected areas only. The community in turn gets rights to all non-

timber forest products and a fixed share from the final timber harvest. Generally, a part of returns from the final timber harvest is used for community development activities, and the remaining part is distributed equally among the members of the community. The agreement is signed between the government representatives and a representative body, known as the Forest Protection Committee (FPC) or Forest Management Committee (FMC) of the community, and the members of FPC or FMC become responsible for the enforcement of the JFM agreement among all the community members. The JFM committee members can hire other members of the community for enforcement purposes. However, the signing of the JFM agreement between the community representatives and the government does not necessarily mean that every member of the community will become a law abider. It is possible that some of the illegal harvesters, before the JFM agreement, may switch over to a group of law abiders just after the JFM agreement, but it is also possible that many members of the community may continue to illegally harvest forest resources even after the JFM agreement. Hence, after the JFM agreement, the members of the community will belong to the groups of cooperators, defectors or enforcers, and their payoffs will be different than the payoffs in non-JFM situations.

1. *Cooperators*: They are the persons in the community who abide by the JFM agreement and do not resort to practices which are illegal under the JFM agreement. In turn, they get a share from the final timber harvest in addition to a proportional share from all the non-timber forest produce.
2. *Defectors*: They are the persons in the community who do not abide by the JFM agreement and resort to illegal removal of forest produce that is not allowed under the JFM. Once they are caught by the enforcers, they are not given any share from the final timber harvest, but they cannot be excluded from community development activities because these activities are of a public nature. Since collection of non-timber forest products is allowed under the JFM agreement, they collect those products as any other member of the community.
3. *Enforcers*: They are the people authorized by the FPC or FMC, who act as watchmen and are responsible for the enforcement of the JFM agreement. They are paid wages for their work.

Therefore, enforcers are the persons in the community who abide by the JFM agreement and are also responsible for enforcing the provisions of the agreement on the community by sanctioning the defectors. Each enforcer receives the same payoff as a cooperator. In addition, he gets a reward from the share of the fines collected from the defectors; however, he has to bear a cost for sanctioning the defectors.

Suppose R_t is the total annual payoff (an annual equivalent) from final timber harvest, R_n is the total annual payoff obtained from non-timber forest produce, R_w is the payoff of the enforcer from annual wages for protection and maintenance of the resource, R_r is the annual payoff of the enforcer from rewards, and R_c is the annual cost of the enforcer in sanctioning the defectors. The annual cost of the enforcer consists of two components—annual fixed costs (F_c) incurred by the enforcer whether he sanctions a defector or not and variable costs (V_c) per unit catch for extra time spent on catching and sanctioning a defector. Suppose s is the share of the community from the final timber harvest, a part of this share, s_1 is used for providing common infrastructural and other community development facilities to the community and the rest s_2 is equally distributed among the cooperators and enforcers. Further suppose that s_c is the proportion of the cooperators population, s_d is the proportion of defectors' population and s_e is the proportion of the enforcers' population, such that $(s_c + s_d + s_e = 1)$. The annual payoff of each type of agent is given by

$$\text{Payoff of a cooperator, } \pi_{c_i} = \frac{s_2 R_t}{(s_c + s_e)n} + \frac{R_n}{n}$$

$$\text{Payoff of a defector, } \pi_{d_i} = \frac{R_n}{n} + (eR_i - bR_f)$$

$$\text{Payoff of an enforcer, } \pi_{e_i} = \pi_{c_i} + \frac{R_f s_d}{s_e} - \left(F_c + \frac{V_c s_d}{s_e} \right) + R_w$$

where $\frac{R_f s_d}{s_e} = R_r$ is the share of reward of each enforcer.

The annual fines for illegal removal, R_f , are assumed to be proportional to the annual payoff from

illegal felling, eR_i , since fines are collected only for the illegal forest produce removed by the defector who is caught by the enforcer, i.e., $(R_f = \alpha eR_i)$, where $\alpha < 1$ for minor thefts not causing a major injury to the resource and do not have an effect on the growth of the resource and $\alpha > 1$ for major thefts causing a major injury to the resource and which have an impact on the growth of the forest resource. The variable cost incurred by the enforcer to sanction a defector, V_c , is also proportional to the payoff from illegal felling, eR_i , i.e., $(V_c = \beta eR_i)$, because if the defector resorts to very high illegal removals of forest produce by exerting a very high effort using sophisticated equipment, the enforcer has to expend a very high cost to catch the defector and so $\beta > 1$. On the other hand, if the defector is using ordinary equipment for illegal removals, does not have any weapons to protect himself and brings the forest produce to the community, the enforcer's cost of catch will be quite low and so $\beta < 1$.

The n -person forest resource use game under JFM regime has many Nash equilibrium, involving credible threats, depending on the parameter values of the payoffs of each type of agent, but it has one *sub-game perfect defection equilibrium* under the following conditions.

1. Payoff of the defector > payoff of the cooperator

$$(eR_i - bR_f) > \left[\frac{s_2 R_t}{(s_c + s_e)n} \right]$$

2. Payoff of the defector > payoff of the enforcer

$$(eR_i - bR_f) > \left[\frac{s_2 R_t}{(s_c + s_e)n} + \frac{R_f s_d}{s_e} - \left(F_c + \frac{V_c s_d}{s_e} \right) + R_w \right]$$

Substituting for $R_f = \alpha eR_i$, in condition (1), we obtain

$$eR_i(1 - b\alpha) > \left[\frac{s_2 R_t}{(s_c + s_e)n} \right]$$

Substituting for $R_f = \alpha eR_i$ and $V_c = \beta eR_i$, in condition (2), we obtain

$$eR_i(1 - b\alpha) > \frac{s_2 R_t}{(s_c + s_e)n} + eR_i(\alpha - \beta) \frac{s_d}{s_e} + (R_w - F_c)$$

These inequalities hold in reality, especially in developing countries where the forest areas are vast

and forest managers (government) have limited resources to protect these areas. Under the prevailing circumstances, the defector knows that the probability of his being caught is low and he exerts a high effort to maximize his payoff from the forest resource. Therefore, the forest resource use game has unique sub-game perfect defection equilibrium, as both these conditions are true in the presence of the following situations

- b is low, the probability of catching a defector is low.
- e is high, the defector applies a high effort for illegal removal of forest produce.
- β is high, the defector is using modern sophisticated equipment for illegal removal of forest produce, whereas the enforcer is not sufficiently equipped to catch the defectors.
- n is high, the population of the forest dependent community is large. It is generally difficult to achieve cooperation in large communities. This result is similar to the findings of [Tyran and Feld \(2002\)](#), who found that conditional cooperation by following other people's behavior is relatively easier in small groups, and [Sally \(1995\)](#) who reported that cooperation is much lower in large populations.
- s_2 is low, share of cooperators from the final timber harvest is low. The cooperators do not have much motivation to save these forests, as they are not getting enough incentives for their cooperative behavior.
- R_w is low, the payoff of enforcer from wages is low. The enforcers are not given sufficient wages for the protection and maintenance of forest resources and they try to find alternate sources of employment for their subsistence needs.
- F_c is high, the enforcer has to spend a high fixed cost in protection of the resource. He is not able to earn enough money from his employment as an enforcer.

Therefore, under the above conditions, no law-breaker is deterred by the threat of punishment and applies a high effort to illegally remove the forest produce from the forest resource leading to defection equilibrium.

However, forest management under JFM regime is not a situation of one time game, but it is a situation of

an infinitely repeated game. In the JFM situation, all the players of the game live in the same community forever, and every member has a chance to observe the strategies and payoff of other members, and learn, select and imitate the successful strategies. Hence, an understanding of evolutionary dynamics of the n -person game is necessary to visualize and develop a full understanding of the outcome of JFM.

5. Evolutionary dynamics of the n -person game under joint forest management regime

In the game situations where the players learn and select successful strategies, the proportion of players choosing a particular strategy evolves with time, and the proportion of players selecting a particular strategy increases when the payoff to an individual for that strategy exceeds the average, over all the strategies, individual payoff for the whole population. The payoff of the population of cooperators, defectors and enforcers in the community is given by

Payoff of cooperators,

$$\pi_c = \left[\frac{s_2 R_t}{(s_c + s_e)} + R_n \right] s_c$$

Payoff of defectors,

$$\pi_d = [R_n + (eR_i - bR_f)n]s_d$$

Payoff of enforcers,

$$\pi_e = \left[\frac{s_2 R_t}{(s_c + s_e)} + R_n + (R_w - F_c)n \right] s_e + (R_f - V_c)s_d n$$

The standard replicator dynamics is given by

$$\dot{s}_i = s_i(\pi_i - \bar{\pi}), \quad i = c, d, e$$

$\dot{s}_i$ is the rate of change of population share for strategy i . $\bar{\pi}$ is the average payoff in the population as a whole. $\bar{\pi} = \sum_i s_i \pi_i$.

An equilibrium or a steady state of the system is a state at which $\dot{s}_i = 0$, i.e., there is no further shift in population composition. A necessary and sufficient condition for a state to be an equilibrium point of the dynamics is that all surviving strategies earn equal

payoffs. Since $(s_c + s_d + s_e) = 1$, the two-dimensional replicator dynamics are

$$\dot{s}_c = s_c(\pi_c - \bar{\pi})$$

$$\dot{s}_d = s_d(\pi_d - \bar{\pi})$$

The possible equilibriums of this replicator dynamics are

1. C equilibrium, in which there are only cooperators.
2. D equilibrium, in which there are only defectors.
3. E equilibrium, in which there are only enforcers.
4. C–D equilibrium, in which there are no enforcers.
5. D–E equilibrium, in which there are no cooperators.
6. C–E equilibrium, in which there are no defectors.

Generally, E equilibrium and C–D equilibrium are not possible because there are always some, but only some and not all, enforcers under this regime, which is a mandatory provision of the JFM agreements. The proof of asymptotic stability for the rest of four equilibriums is given in Appendix A, and implications of the conditions under which these equilibriums are asymptotically stable are discussed next.

5.1. Stability of the C equilibrium

Cooperation equilibrium is an equilibrium in which there are only cooperators in the population and no defectors or enforcers. The cooperation equilibrium is asymptotically stable only as long as payoff of the cooperators is positive (for proof, see Appendix A). However, the stability of the cooperation equilibrium is based on a very strong assumption that there are no defectors or enforcers in the population. In fact, it is almost impossible to ensure a pure population of cooperators because, even if a single person defects in the community, he earns a higher payoff as there is nobody to sanction him and the entire equilibrium will shift to defection equilibrium. In addition, there will always be some enforcers as per the provision of JFM agreement. Therefore, in reality, it is not possible to have pure stable cooperation equilibrium, without the presence of some enforcers. We need to have some population of enforcers, howsoever small it may be, to sanction any possible future defections.

5.2. Stability of the D equilibrium

Defection equilibrium is an equilibrium in which there are only defectors in the population and no cooperators or enforcers. The defection equilibrium is asymptotically stable only if b and α are low and β is high (for proof, see Appendix A). The probability of catching, b , is usually low when there are no enforcers to catch and sanction the defectors. A low value of α means that the fines imposed for illegal removal of forest produce are not high and the defectors are not deterred by these fines. They apply a very high effort so as to maximize their payoffs and make the defection equilibrium asymptotically stable. It is much easier to achieve such equilibrium if the defectors are equipped with sophisticated equipment (high β). Sethi and Somanathan (1996) have proven that D equilibrium is asymptotically stable in the use of common pool fisheries resource for all parameter values, because defectors perform better than cooperators in cases there are no enforcer to sanction the defectors. However, the defection equilibrium is not asymptotically stable if the government appoints a few enforcers and provides them with sufficient equipment to catch and sanction the defectors.

5.3. Stability of the D–E equilibrium

D–E equilibrium is an equilibrium in which there are only defectors and enforcers and no cooperators. The D–E equilibrium is asymptotically stable only if $\pi_c < \pi_d$ (for proof, see Appendix A). However, the D–E equilibrium will never be stable if the enforcers' payoff is higher than defectors' payoff. If the enforcers are given high rewards for sanctioning the defectors in addition to their wages and they are sufficiently equipped to catch the defectors, their payoff will always be higher than those of defectors. In the presence of such motivated enforcers, the probability of catch substantially increases and the defectors' payoff decreases further. Since the enforcers also get a share from the non-timber forest produce and from the final timber harvest, the defectors can never have a higher payoff than the enforcers and as such it is impossible to have only defectors and enforcers in an asymptotically stable equilibrium state. However, it is not impossible that the enforcers' payoffs are set, either by the Forest Department or by Forest Protection

Committee, at quite low level, and in such cases D–E equilibrium will be stable.

5.4. Stability of the C–E equilibrium

C–E equilibrium is an equilibrium in which there are only cooperators and enforcers and no defectors. The C–E equilibrium is asymptotically stable if s_2 , R_t , R_n and R_w are high, and F_c is low (for proof, see Appendix A). These conditions imply that, over a period of time, community forest resource use under the JFM regime will move towards the stable equilibrium of cooperators and enforcers if (i) the individual share of the members of the community from the net final timber harvest, s_2 is high; (ii) the net payoff from timber and non-timber harvest of the resource, R_t and R_n are high. This is possible by ensuring a high price of timber and non-timber products from the resource. In addition, both R_t and R_n depend on site quality and age of the forests; therefore, the stability of the C–E equilibrium also depends on these factors. The C–E equilibrium is stable if either the site quality improves or, for a particular site quality, the forests are allowed to grow up to maturity. Since the local communities are dependent on forest resources for their subsistence needs, their behavior is guided by both the short-term needs and long-term benefits. A good price and market for non-timber forest produce provides the short-term security and a high price of the final timber harvest ensures the long-term benefits to the community. (iii) The wages provided by the government for protection and maintenance of the resource, R_w , are high and the fixed cost to guard the forest resource, F_c , is low and the enforcer is suitably compensated by the wages for his work of protection and maintenance. This will help in effective monitoring of the resource by the enforcers and subsequently the costs of defection will be high as compared to the benefits derived from self-interested behavior of resource exploitation by the defectors. Under these conditions, if there exists a state which is close to stable C–E equilibrium with a few defectors, then the population will be driven asymptotically under evolutionary pressures into this stable state.

Consequently, there are four possible evolutionary strategies equilibria—C–equilibrium, D equilibrium, D–E equilibrium and C–E equilibrium—of the n -person forest resource use game under JFM regime,

however; C equilibrium and D equilibrium are almost impossible given the institutional arrangements of JFM, and D–E and C–E can be asymptotically stable under different scenarios.

6. Summary and conclusion

In this paper, we analyzed the behavioral strategies of members of forest-based communities in the use of a state owned forest resources, considering these communities to be heterogeneous in character. We divided the entire population into three different groups—cooperators, defectors and enforcers—based on the strategies they follow in using these forest resources. We found that for a forest under state regime, the resource use game has a unique Nash equilibrium in which defectors earn a higher payoff and they exploit the forest resource, leading to its degradation and even complete extinction. However, the forest resource use game under JFM regime has many Nash equilibria, but has unique sub-game perfect defection equilibrium. We also studied the evolutionary dynamics of the game under JFM regime, and found that this game has four evolutionary strategies equilibria—cooperators (C) equilibrium, defectors (D) equilibrium, defectors–enforcers (D–E) equilibrium and cooperators–enforcers (C–E) equilibrium, but it has only two asymptotically stable equilibria, C–E and D–E.

These results are very important from JFM policy prescription and management perspective. The results indicate that the success of JFM is not automatic, but it will depend upon policy prescriptions and their relevance to the conditions of forests and communities. As demonstrated above, the long-run outcome of the program may be either D–E equilibrium or C–E equilibrium. However, in the short-run, any of the outcomes is possible which explains spatial and temporal variability in the outcomes of JFM reported in the literature.

The necessary conditions for the stability of D–E and C–E equilibria can be used by policymakers and resource managers in the implementation of JFM program. Forest Departments and FPCs have to ensure that the payoffs of enforcers are good enough to make sure that D–E equilibrium is not stable. Similarly, to make sure the stability of C–E equilibrium, the FD has to devise appropriate policies. For example, first, the government has to ensure a market for the non-timber forest produce, in addition to providing a high price for

both the timber and non-timber forest products. Second, the communities have to be provided with a substantial share of the final timber harvest in order to sustain the JFM program. Third, the enforcers should be provided with higher wages than their cost for protection and maintenance of the resource. The enforcers should also be provided with sufficient equipment so as to enforce the cooperative norms on the defectors, who are usually better equipped.

In conclusion, the government should pay very careful attention to the conditions worked out in this paper and frame its policies and strategies accordingly in order to manage these forest resources in a sustainable way. However, one limitation of our analysis is that we have assumed that the annual payoffs to members of local communities remain the same over different years, which may not be realistic specifically in the context of forest resources, because of their natural growth. Hence, further research in evolutionary game-theoretic formulation of JFM should include the resource stock dynamics, where the payoffs from the forest resource vary with age.

In addition, evolutionary game theory is also based on an assumption that an individual is procedurally rational. Even though procedural rationality, the way it is defined in evolutionary game theory literature, allows imitation and learning, but an individual is still a selfish person. The current evolutionary game theory formulations do not allow an individual to have “other-regarding preferences” or “social preferences” while in village communities of India these preferences can be observed quite commonly. Hence, game-theoretic formulations of forest management regimes have to be extended beyond the current structure of evolutionary game theory.

Appendix A

A.1. Stability of the C equilibrium

In order to test the asymptotic stability of the cooperation equilibrium, we use the Jacobian of the two dimensional system.

$$J = \begin{bmatrix} \frac{\partial \dot{s}_c}{\partial s_c} & \frac{\partial \dot{s}_c}{\partial s_d} \\ \frac{\partial \dot{s}_d}{\partial s_c} & \frac{\partial \dot{s}_d}{\partial s_d} \end{bmatrix}$$

$$J = \begin{bmatrix} (\pi_c - \bar{\pi}) + s_c \frac{\partial (\pi_c - \bar{\pi})}{\partial s_c} & s_c \frac{\partial (\pi_c - \bar{\pi})}{\partial s_d} \\ s_d \frac{\partial (\pi_d - \bar{\pi})}{\partial s_c} & (\pi_d - \bar{\pi}) + s_d \frac{\partial (\pi_d - \bar{\pi})}{\partial s_d} \end{bmatrix}$$

Cooperation equilibrium is an equilibrium in which there are only cooperators, $s_c = 1$, and no defectors and enforcers, $s_d = s_e = 0$. The average payoff of the community $\bar{\pi} = \pi_c$.

The payoffs of each group of agents are

$$\begin{aligned} \pi_c &= \left[\frac{s_2 R_t}{s_c} + R_n \right] s_c \\ \pi_d &= 0 \\ \pi_e &= 0 \end{aligned}$$

The elements of the Jacobian are

$$\begin{aligned} J_{11} &= -\pi_c < 0 \\ J_{12} &= 0 \\ J_{21} &= 0 \\ J_{22} &= -\pi_c < 0 \end{aligned}$$

For local asymptotic stability of the cooperation equilibrium, the determinant of the Jacobian has to be positive and its trace negative. The determinant of this Jacobian is positive and its trace is negative if $\pi_c > 0$. Therefore, the cooperation equilibrium is asymptotically stable if $\pi_c > 0$.

A.2. Stability of the D equilibrium

Defection equilibrium is an equilibrium in which there are only defectors, $s_d = 1$, and no cooperators and enforcers, $s_c = s_e = 0$. The average payoff of the community $\bar{\pi} = \pi_d$.

The payoffs of each group of agents are

$$\begin{aligned} \pi_c &= 0 \\ \pi_d &= [R_n + (eR_i - bR_f)n] \\ \pi_e &= (R_f - V_c)n \end{aligned}$$

The elements of the Jacobian are

$$\begin{aligned} J_{11} &= \pi_c - \bar{\pi} = -\pi_d < 0 \\ J_{12} &= 0 \\ J_{21} &= \pi_e \\ J_{22} &= \pi_e - \pi_d \end{aligned}$$

For local asymptotic stability of the defection equilibrium, the determinant of the Jacobian has to be positive and its trace negative. For these conditions to be satisfied, we should have $\pi_d > \pi_e$.

$$\Rightarrow R_n + (eR_i - bR_f)n > (R_f - V_c)n$$

Assuming payoff from NTFP, R_n , to be small as compared to payoff from illegal removal of timber, $(eR_i - bR_f)$, the modified condition becomes

$$(eR_i - bR_f) > (R_f - V_c)$$

Substituting for $R_f = \alpha eR_i$ and $V_c = \beta eR_i$, the simplified condition is

$$(1 - b\alpha) > (\alpha - \beta)$$

Therefore, the defection equilibrium is asymptotically stable if b and α are low and β is high.

A.3. Stability of the D–E equilibrium

D–E equilibrium is an equilibrium in which there are only defectors and enforcers, $(s_d + s_e) = 1$, and no cooperators, $s_c = 0$. The average payoff of the community $\bar{\pi} = \pi_d$ or π_e .

The payoffs of each group of agents are

$$\pi_c = 0$$

$$\pi_d = l(eR_i - bR_f)n | s_d$$

$$\pi_e = \left[\frac{s_2 R_t}{(s_c + s_e)} + R_n + (R_w - F_c)n \right] s_e + (R_f - V_c)s_d n$$

The elements of the Jacobian are

$$J_{11} = -\pi_d < 0$$

$$J_{12} = 0$$

$$J_{21} = s_d \pi_e$$

$$J_{22} = (\pi_e - \pi_d)s_d < 0 \text{ if } \pi_e < \pi_d$$

For local asymptotic stability of the D–E equilibrium, the determinant of the Jacobian has to be positive and its trace negative. The determinant of this Jacobian is positive and its trace negative if $\pi_e = \pi_c = \pi_e$. Hence,

the D–E equilibrium is asymptotically stable only if $\pi_e < \pi_d$.

A.4. Stability of the C–E equilibrium

In the C–E equilibrium, there are only cooperators and enforcers, $(s_c + s_e) = 1$, and no defectors, $s_d = 0$. The average payoff of the community $\bar{\pi} = \pi_c = \pi_e$. The payoffs of each group of agents are:

$$\pi_c = \left[\frac{s_2 R_t}{(s_c + s_e)} + R_n \right] s_c$$

$$\pi_d = 0 \quad \pi_e = \left[\frac{s_2 R_t}{(s_c + s_e)} + R_n + (R_w - F_c)n \right] s_e$$

The elements of the Jacobian are

$$J_{11} = 0$$

$$J_{12} = s_c \pi_e$$

$$J_{21} = 0$$

$$J_{22} = \pi_d - \bar{\pi} = -\pi_e < 0$$

The determinant of this Jacobian is 0 and for local asymptotic stability of the C–E equilibrium, the trace of the Jacobian has to be negative, which requires $\pi_e > 0$ or

$$\left[\frac{s_2 R_t}{(s_c + s_e)} + R_n + R_w n \right] > [F_c n]$$

Therefore, the C–E equilibrium is asymptotically stable, if s_2 , R_t , R_n and R_w are high, and F_c is low.

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